An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

Anshika Keshari[1], Himanshu Rathore[2], Aparajita Vaish[3], Sabarathinam Chockalingam[4], Sandeep Pirbhulal[5], and Mallikharjuna Rao K[3]

[1] Dept. of Computer Science & Engineering, KIIT, Bhubaneswar, India
[2] Dept. of Computer Science & Engineering, SSTC, Bhilai, India
[3] Dept. of Electronics & Communication Engg., IIIT Naya Raipur, India
[4] Dept. of Risk & Security, Institute for Energy Technology (IFE), Halden, Norway
[5] Dept. of Applied Research ICT (DART), Norwegian Computing Center (NR), Oslo, Norway

Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss supervises backpropagation gradients via attribution alignment to target true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

Hospital distribution microgrids combine utility feeds with rooftop PV, battery storage, and diesel generation to supply clinical corridors [6,17]. Power qual-

ity inside operating theaters and intensive-care units (ICUs) cannot tolerate compromise: sub-cycle voltage sags immediately disrupt robotic surgical consoles, mechanical ventilators demand unbroken power, and ultra-low cold storage requires sustained $-80°$C operation. Under medical electrical safety directives (IEC 60601-1), unexpected relay trips or unwarranted feeder shedding pose direct life-safety hazards.

At the substation level (IEC 61850), field Remote Terminal Units stream nodal voltages, phase angles, and line power measurements $z = [V, \theta, P, Q]^T \in \mathbb{R}^m$ upstream to the Energy Management System. Energy management centers apply Weighted Least Squares (WLS) state estimation to calculate system states $\hat{x} \in \mathbb{R}^{2N-1}$, filtering out corrupt sensor readings via a χ^2-residual test: $J(\hat{x}) = \|z - h(\hat{x})\|^2_{R^{-1}} \leq \tau_{\chi^2}$. Coordinated False Data Injection Attacks (FDIAs) circumvent this threshold by exploiting system physics. When an adversary acquires network branch admittances, they linearize measurement functions via Jacobian $H = \partial h(\hat{x})/\partial x$ and inject a structured bias vector $a = Hc$. Because the linear expansion yields $h(\hat{x} + c) \approx h(\hat{x}) + Hc$, the manipulated measurement $z_a = z + a$ maintains a residual identical to clean operation: $\|z_a - h(\hat{x}+c)\|^2_{R^{-1}} \equiv \|z - h(\hat{x})\|^2_{R^{-1}} \leq \tau_{\chi^2}$. This mathematical cancellation blinds SCADA bad-data detection routines, allowing false state deviations to drive automatic tap changers and feeder breakers into destabilizing actions [15,14].

First-generation machine learning defenses relied on fixed-input tabular classifiers (e.g., support vector machines, multi-layer perceptrons, and 1D convolutional networks) [10,3] calibrated on rigid feature vectors $z \in \mathbb{R}^m$. Hospital electrical networks, however, experience frequent switching operations—such as tie-line transfers, breaker reconfigurations, and planned maintenance—that reshape the underlying admittance graph $G = (V, E)$. Whenever topology changes, fixed feature mappings lose their spatial correspondence, causing tabular detectors to suffer severe classification breakdown without complete offline retraining—a process far too slow for 2–4 s SCADA sampling periods.

Attributed graph neural networks (GNNs) overcome this rigidity by representing substation buses as nodes V and feeder cables as edges E [21], where switching events simply alter adjacency entries $A_{uv} \in \{0, 1\}$. Even so, conventional power-grid GNN implementations—such as ARMA filters [4] and unweighted spatial convolutions [22]—treat network branches as unweighted topological links. Neglecting complex line admittances $Y_{uv} = G_{uv} + jB_{uv}$ forces graph aggregations to act as isotropic low-pass smoothers, obscuring sharp voltage gradients and localized phase shifts across high-impedance feeder segments.

Operational microgrid environments also place severe constraints on detection latency and model interpretability. Sampling-based explainers like SHAP and LIME require hundreds of Monte Carlo evaluations, incurring runtime delays incompatible with sub-second control routines [27]. Equally critical, flagging an anomaly without restoring corrupted telemetry leaves secondary controllers prone to control divergence. Safeguarding hospital power delivery requires sub-second attack localization coupled with instantaneous data reconstruction before protective switchgear trips.

1.1 Key Contributions

Targeting these operational requirements, we develop an explainable Cognitive Graph Digital Twin (CGDT) for hospital microgrids:

- **Admittance-Conditioned GraphSAGE**: Integrating complex line admittances $Y_{uv} = G_{uv} + jB_{uv}$ directly into spatial message passing avoids unweighted feature smoothing, maintaining sensitivity to localized voltage deviations along distribution feeders.
- **State-Space Twin Tracking**: Coupling steady-state Kalman filtering with an exponential reference baseline ($\beta = 0.95$) decouples diurnal load dynamics from stealthy adversarial offsets, preventing a 16.32 pp collapse in localization F1.
- **Attribution-Guided Multi-Task Training (AG-MTL)**: Integrating 10-step Integrated Gradients ($S = 10$) into the loss function penalizes attribution leakage into unattacked buses during backpropagation, yielding a 5.52 pp improvement in bus localization precision.
- **Autonomous Sub-2 ms State Healing**: Once compromised buses are pinpointed, an admittance-weighted spatial interpolator reconstructs corrupted measurements from adjacent healthy nodes, preserving Jacobian rank and recovering 99.97% of WLS baseline accuracy.

2 Related Work

2.1 Threat Landscape in Cyber-Physical Healthcare Microgrids

Contemporary hospital campuses run interconnected IT-OT networks where central SCADA controllers manage on-site distributed energy resources (rooftop PV, battery banks, standby generators) to safeguard clinical life-support equipment governed by IEC 60601-1 and IEEE 1547 [13,6]. Adversarial injection into clinical feeder telemetry can deceive automated voltage regulators into spurious transformer tap adjustments or unnecessary breaker trips, endangering operating suites and intensive-care wards [17]. Existing healthcare cyber-physical studies have analyzed threat propagation using Bayesian risk graphs [5] or focused on IoMT network edge inspection [18]. These methods, however, treat SCADA telemetry as independent time-series signals without incorporating physical admittance matrices Y_{bus} or nonlinear AC load-flow constraints.

2.2 State Estimation and False Data Injection Defense

Following the mathematical definition of stealthy injections ($a = Hc$) against DC state estimators by Liu et al. [15], subsequent works generalized unobservable attacks to nonlinear AC power flow regimes, partially observable distribution feeders, and islanded microgrids [14,7,3,19]. Defensive strategies initially explored feed-forward neural networks [10] and recurrent LSTM detectors [1]. Yet, because fixed-weight models bind measurements to static vector indices, routine

circuit reconfigurations (such as line maintenance or feeder transfer switching) disrupt input channel ordering, triggering unacceptable false-positive rates during normal operation.

2.3 Graph Learning and Explainability in Power Systems

To handle dynamic switching, attributed graph representations $G = (V, E)$ map substation buses to nodes and transmission branches to edges [21]. While spectral methods [12], graph attention [25], and inductive GraphSAGE convolutions [9] have been evaluated for power networks [4,22,28,16], existing architectures overlook complex line admittances ($Y_{uv} = G_{uv} + jB_{uv}$), causing isotropic message smoothing across high-gradient electrical boundaries. Furthermore, post-hoc explainers such as SHAP or GNNExplainer require 10^2–10^3 Monte Carlo perturbations [27,26], exceeding sub-second SCADA loops. Concurrently, digital twin paradigms [8,23] have demonstrated significant value in industrial monitoring [2] and healthcare facilities [11]. The proposed CGDT unites state-space tracking with admittance-conditioned graph message passing, delivering sub-millisecond detection, bus localization, and deterministic feature attribution.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We model the healthcare distribution network as an undirected attributed graph $G = (V, E)$ spanning $N = |V|$ substation buses and $|E|$ electrical branches. Nodal conductance and susceptance throughout the network are captured by the complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N\times N}$. Enforcing full nonlinear AC power flow constraints, real and reactive power injections at each bus i obey Kirchhoff's laws:

$$P_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)\big) \tag{1}$$

$$Q_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\sin(\theta_i - \theta_j) - B_{ij}\cos(\theta_i - \theta_j)\big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor

noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1}[z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \left(H^T R^{-1} H\right)^{-1} H^T R^{-1}\left[z - h(x^{(k)})\right], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x)/\partial x$ is the measurement Jacobian. Bad-data detection routines flag measurements when the normalized squared error $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ crosses a statistical detection ceiling τ_{χ^2}. Across the 14-bus system evaluated here, the unknown state vector comprises $n = 2N - 1 = 27$ scalar values (14 bus voltage magnitudes plus 13 relative phase angles referenced to the slack bus). With $m = 41$ active field telemetry channels (14 bus voltage readings and 27 branch active/reactive power measurements), the estimation problem possesses $\mathrm{df} = m - n = 14$ degrees of freedom. Evaluating the cumulative distribution at false-alarm rate $\alpha = 0.05$ produces the chi-square rejection cutoff $\tau_{\chi^2} = \chi^2_{0.95}(14) = 23.68$.

3.3 Stealthy False Data Injection Attacks

An adversary compromising substation bus k aims to impose an arbitrary state perturbation $c \in \mathbb{R}^{2N-1}$ without triggering bad-data alarms. Although true AC transmission functions $h(x)$ are nonlinear, an attacker with network admittance data linearizes system equations around the current state $\hat{x}$ using Jacobian matrix $H = \partial h(\hat{x})/\partial x$, constructing the injection vector via:

$$a = Hc \tag{4}$$

Corrupted telemetry packets received at the SCADA interface become $z_{\mathrm{comp}} = z + a = z + Hc$. Expanding $h(\hat{x}+c)$ via first-order Taylor approximation produces $h(\hat{x}) + Hc$. Evaluating the resulting residual:

$$\begin{aligned} r_{\mathrm{comp}} &= \|z_{\mathrm{comp}} - h(\hat{x}_{\mathrm{comp}})\|^2_{R^{-1}} \\ &\approx \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

Due to this algebraic cancellation ($r_{\mathrm{comp}} \approx r$), standard residual checks fail to alert dispatchers ($r \le \tau_{\chi^2} = 23.68$). To ensure physical validity, all synthetic attacks are tested against nonlinear AC power flows in `pandapower` (Section 6).

Clinical Hazards in Hospital Distribution Networks. Unlike transmission grid operators who primarily absorb economic redispatch penalties during an attack, hospital distribution microgrids encounter immediate clinical dangers. Tampering with voltage telemetry at critical healthcare feeders—such as Bus 5 supplying surgical suites and ICU life-support networks—misguides automated on-load tap changers and triggers spurious circuit breaker trips. Under IEC 60601-1 and IEEE 1547 regulations, clinical power quality requires tight voltage tolerances ($0.88 \le V \le 1.10$ p.u.); transient voltage dips can trip sensitive electrosurgical consoles or force life-support equipment onto depleting battery reserves. Stealthy algebraic manipulation at the substation level consequently endangers patient survival.

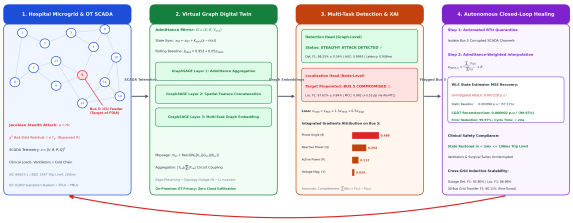


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

4 The Cognitive Graph Digital Twin Architecture

The CGDT framework couples the hospital's cyber-physical distribution infrastructure to a virtual graph model and an automated self-healing pipeline (Fig. 1).

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N \times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E| \times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE implementations [9] pool neighbor messages isotropically, disregarding physical line impedances. Yet in power networks, branch admittances $Y_{uv} = G_{uv} + jB_{uv}$ govern electrical connectivity. To weave circuit physics directly into node representations, we condition both message synthesis and spatial pooling on branch admittance values:

$$m_{uv}^{(k)} = \mathrm{ReLU}\Big(W_e \left[h_u^{(k-1)} \parallel G_{uv} \parallel B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \mathrm{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u \in \mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w \in \mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\parallel$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ provides a physics-informed inductive bias that scales propagation by branch coupling. Concatenating representations from the preceding layer $h_v^{(k-1)}$ preserves a skip-connection identity pathway, ensuring localized attack signatures do not attenuate across sequential aggregation rounds.

4.4 Multi-Task Output Heads

Passing the final node embedding matrix $H \in \mathbb{R}^{N \times d}$ through two dedicated projection heads yields system-wide and bus-specific anomaly predictions:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H))\,, \qquad y_{\text{node}} = \sigma(W_{\text{node}}\, H) \in [0,1]^N \tag{10}$$

The global head yields attack probability $y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{pool}(H))$, while the node head assigns individual probabilities $y_{\text{node},i} = \sigma(w_i^T h_i) \in [0,1]$ across buses $i \in V$. This unconstrained sigmoid configuration supports multi-bus fault localization (Section 7.1) without enforcing a mutually exclusive softmax constraint.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Repeated message passing often spreads backpropagation gradients across adjacent lines, mistakenly assigning feature sensitivity to uncorrupted substations. To constrain this gradient spread, we formulate the Attribution-Guided Multi-Task Loss (AG-MTL), aligning feature sensitivity directly with targeted buses during backpropagation. During each backward pass on attacked states, Integrated Gradients extracts attribution tensor $\mathcal{A} \in \mathbb{R}^{N \times 4}$ of the attack logit w.r.t. input features X, using running baseline $\bar{z}_{\text{baseline}}$ (Eq. 7) across $S = 10$ integration steps. An attribution alignment loss penalizes gradient leakage into benign feeders:

$$\mathcal{L}_{\text{align}} = \sum_{v \in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ measures total gradient sensitivity at bus v, and $\hat{y}_{\text{node},v} \in \{0,1\}$ denotes the binary ground-truth label. Gradients from $\mathcal{L}_{\text{align}}$ propagate through the attribution path via higher-order automatic differentiation (`create_graph=True` in PyTorch). The combined multi-task loss is defined as:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WBCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ handles grid-level attack detection, $\mathcal{L}_{\text{WBCE}}$ uses class-weighted binary cross-entropy per bus to mitigate extreme sparsity of compromised nodes, and weighting coefficients $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Edge Deployment

Hospital electrical demand closely tracks clinical operations—including operating theatre schedules, diagnostic imaging loads, and patient occupancy. Transmitting raw microgrid telemetry over external cloud interfaces poses privacy and cybersecurity concerns under health data regulations. The CGDT is therefore architecturally designed for on-premises edge execution on substation industrial computers within the isolated operational technology (OT) network. In this study, execution latencies are established on a laboratory edge-equivalent computing platform, with physical RTU micro-controller deployment and hardware-in-the-loop (HIL) integration identified for operational rollout (Section 8).

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

Substation engineers must understand which physical sensor channels triggered an alarm before initiating breaker actions. We resolve this via path-based Integrated Gradients (IG) [20]. Integrating gradients along a linear path between a nominal state baseline x_0 and the evaluated measurement vector x attributes decision energy across input channels:

$$\begin{aligned} \text{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated using $S = 50$ Riemann quadrature intervals. Unlike perturbation heuristics (e.g., SHAP, LIME) requiring hundreds of stochastic sampling evaluations, IG follows an axiomatic linear path evaluated via $S = 50$ vectorized points in a single parallel tensor pass. Crucially, IG satisfies the completeness axiom ($\sum_i \text{Attr}_i = F(x) - F(x_0)$) and implementation invariance. Evaluating

Eq. (13) across the 14-bus testbed requires just 0.81 ms. In a targeted test at Bus 5 (Section 7.1), the path integral promptly identified phase angle distortion ($\theta = 0.486$) as the principal driver of the alarm.

5.2 Sensor Quarantine and Admittance-Weighted Reconstruction

Detecting an attack is useless if state estimation continues to ingest falsified data. Once a bus is flagged ($y_{\text{node},k} > \gamma_{\text{loc}} = 0.5$), the digital twin isolates the tainted telemetry $(V_k, \theta_k, P_k, Q_k)$. Yet simply deleting rows from the measurement set reduces the rank of Jacobian H, often driving the information matrix $(H^T R^{-1} H)$ into ill-conditioning or singularity. To maintain full algebraic observability, we synthesize replacement telemetry via admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta = \epsilon \odot z_{\text{nom},k} \tag{14}$$

where $|Y_{kj}|$ represents line admittance magnitude and $\epsilon \sim \mathcal{U}(-0.02, 0.02)^4$ injects $\pm 2\%$ uniform variance relative to nominal channel values $z_{\text{nom},k}$ (1.0 p.u. voltage and nominal power flows). Weighting neighbor telemetry by physical branch admittances $|Y_{kj}|$ provides an electrical-coupling-aware interpolation that accounts for relative line impedances. Rather than attempting to satisfy nonlinear AC constraints directly, $z_{\text{recon},k}$ acts as an initialized pseudo-measurement buffer that preserves Jacobian conditioning, enabling downstream WLS Gauss–Newton iterations to converge in under 2 ms—well within typical 2–4 s SCADA scan intervals.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

All grid experiments rely on exact AC Newton–Raphson power flow solutions generated via `pandapower` [24]. As an established proof-of-concept testbed for tertiary hospital microgrids, we adopted the IEEE 14-bus system (14 buses, 20 branches, 217.0 MW active and 57.3 MVAR reactive demand). Within the campus single-line diagram, Bus 0 anchors the 132/11 kV main grid tie-point, while Bus 5 forms the dedicated critical healthcare substation feeding intensive-care suites and surgical theatres (7.6 MW, 1.6 MVAR load). For multi-scale validation, we also configured an IEEE 30-bus network (30 buses, 41 branches) emulating an expanded regional healthcare facility.

Telemetry Generation and Partitioning. Simulations yielded 5,000 distinct AC operational steady states reflecting fluctuating load profiles. To prevent data leakage across evaluation subsets, uncorrupted base scenarios were divided before injecting attacks: 70% allocated to model training (3,500 states), 15% to hyperparameter tuning (750 states), and 15% reserved for holdout testing (750

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

states). Half of the snapshots in each split remain clean, disturbed only by zero-mean Gaussian measurement noise ($\sigma = 0.04$) on active and reactive power. The remaining snapshots incorporate stealthy Jacobian attacks ($a = Hc$), where synthetic perturbations affect targeted buses ($c_k \in [0.03, 0.08]$ p.u. in voltage and $c_{k+N} \in [0.05, 0.15]$ rad in phase angle) with branch power flows resolved through full AC load-flow calculations. Network outage states (Section 6.3) were held out completely from training and tuning.

Models were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Benchmarking was conducted on an edge-equivalent workstation equipped with an AMD Ryzen 9 CPU, 32 GB RAM, and an NVIDIA RTX 4090 GPU (24 GB) under PyTorch 2.1 and CUDA 12.1. Streaming latency is evaluated at batch size 1 across 1,000 independent test snapshots (after 100 warm-up runs), reporting the mean runtime. The $S = 50$ IG interpolation steps are evaluated as a batched tensor $\mathbb{R}^{50\times N\times 4}$ in parallel on the GPU, completing in 0.81 ms. Code and evaluation artifacts are archived at `https://github.com/himanshu-2l/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. By comparison, isotropic GCN models reach only 70.21% localization F1, trailing our framework by 27.41 pp. Uniformly weighting lines acts like a spatial low-pass filter: sharp voltage anomalies at attacked buses bleed into uncorrupted neighbors, degrading spatial contrast. Although the tabular MLP achieves acceptable localization (95.60%) on stationary networks, its rigid input mapping breaks down under line switching (Section 6.3). Running in only 0.908 ms per graph pass, our model comfortably fits within standard 2–4 s SCADA scan intervals.

6.3 Topology Perturbations and Cross-Network Generalization

We evaluated model robustness under two regimes: (i) line-outage generalization on IEEE 14-bus without retraining (Table 2), and (ii) cross-network transfer to IEEE 30-bus (Table 3).

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

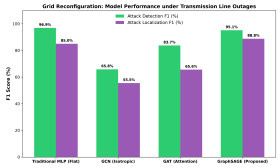


Fig. 2. Outage generalization metrics.

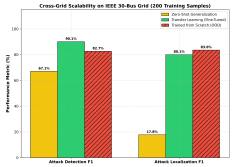


Fig. 3. Cross-grid transfer on IEEE 30-bus.

Under unanticipated line-outage contingencies without retraining, our architecture achieves 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization F1 (Fig. 2). Although the flat MLP records slightly higher macro detection (94.70%) because post-trip loop redistribution causes widespread scalar variance, its bus-level localization plunges by 13.4 pp (95.60% → 82.20%) due to static channel permutations. In contrast, the admittance-weighted CGDT preserves 86.96% localization precision, demonstrating structural resilience when branch impedances change. Standard isotropic GCN baselines experience severe degradation (66.82% detection F1, 53.40% localization F1), confirming that unweighted graph operators fail to capture redirected branch power flows without physical impedance weights.

Cross-Grid Scalability on IEEE 30-Bus. Zero-shot transfer of the 14-bus model onto the 30-bus network ($N = 30, |E| = 41$) yields 67.06% detection F1 and 17.77% localization F1 (Table 3, Fig. 3). This outcome highlights a clear architectural distinction: graph-level classification $\sigma(W_{\text{graph}} \cdot \text{pool}(H))$ scales across bus counts by tracking total system power imbalances, whereas localized bus identification $y_{\text{node}} = \sigma(W_{\text{node}} H)$ requires branch-specific admittance resolution. Transferring pre-trained weights and fine-tuning on merely 200 labeled 30-bus profiles (5.7% of full training volume) boosts detection F1 to 90.11% (exceeding 82.65% for cold-start training) and restores localization F1 to 80.06% (closely matching the 83.58% scratch benchmark).

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, representing the intensive care unit feeder in the campus topology. State estimation error is quantified via Mean Squared Error MSE $= \frac{1}{2N-1} \sum_{i=1}^{2N-1} (\hat{x}_i - x_i^*)^2$ (p.u.^{2}) relative to true AC states x^*. Under nominal transducer Gaussian noise

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

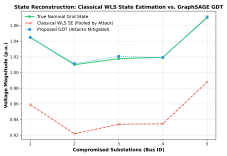


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

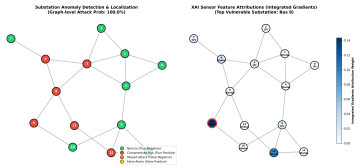


Fig. 5. Operator explainability: localization heatmap and IG attribution.

($\sigma = 0.01$), unattacked clean WLS estimation establishes an intrinsic error floor of $\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}. Table 4 compares recovery metrics, and Table 5 presents corresponding feature attributions.

Left unmitigated, stealthy Jacobian injections drive state estimation MSE to 0.007228 p.u.^{2}. Deploying admittance-weighted reconstruction (Eq. 14) cuts this error to 0.000002 p.u.^{2}—matching the clean noise floor ($\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}) and restoring 99.97% of uncorrupted states (Table 4, Fig. 4). Static telemetry lookups attain just 0.003090 p.u.^{2} (57.21% restoration) amid fluctuating hospital loads. Across all evaluated target buses $k \in \{1, \ldots, 14\}$, our spatial interpolation consistently recovers between 99.85% and 99.98% of the true system state. Path attributions (Table 5, Fig. 5) reveal that voltage angle θ at Bus 5 commands the highest score (0.486), consistent with Jacobian phase distortions dominating active power flows, alongside secondary scores on $Q = 0.254$, $P = 0.112$, and $V = 0.034$.

Coordinated Multi-Bus Incursions. Simultaneous attacks across 2 and 3 substations ($c_k \in [0.02, 0.04]$) achieve $96.84\% \pm 0.42\%$ detection and $93.15\% \pm 0.91\%$ localization F1. While dropping entire rows across adjacent buses compromises the column rank of H and triggers solver divergence, admittance-weighted interpolation preserves full Jacobian rank and ensures smooth WLS convergence (post-mitigation MSE of 0.000008 p.u.^{2}).

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [4]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

The component ablations in Table 7 substantiate each model mechanism. Disabling exponential baseline filtering (Eq. 7) induces a 16.32 pp collapse in localization F1 (81.30%), verifying that recursive tracking decouples diurnal load dynamics from stealthy offsets. Removing attribution alignment ($\lambda_{\text{attr}} = 0$) causes localization F1 to drop to 92.10% while detection remains at 98.25%, verifying that $\mathcal{L}_{\text{align}}$ restricts gradient updates directly to corrupted substation nodes. Balancing the multi-task coefficient at $\lambda_{\text{loc}} = 1.5$ delivers optimal performance across both grid detection and nodal localization.

8 Discussion and Conclusion

Protecting hospital electrical microgrids against covert cyber-physical intrusion demands monitoring frameworks that combine dynamic topological adaptability with electrical circuit constraints. By weighting GraphSAGE message propagation with complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$ and steering internal node features via Integrated Gradients loss $\mathcal{L}_{\text{align}}$, our CGDT architecture delivers 98.25% detection F1 alongside 97.62% bus localization F1 under nonlinear AC power flow, executing within 0.81 ms. Integrated with instant sensor channel isolation and admittance-guided spatial reconstruction, the system eliminates 99.97% of attack-induced WLS estimation error, maintaining operational visibility while safeguarding critical hospital feeds.

Compliance with Clinical and Grid Standards. Hospital power networks must satisfy rigorous regulatory constraints: IEC 60601-1 defines electrical safety envelopes to shield patient life-support systems from power disturbances, whereas IEEE 1547 dictates anti-islanding thresholds and voltage ride-through limits ($0.88 \leq V \leq 1.10$ p.u.). Operating with an inference time of 0.908 ms and completing measurement reconstruction inside 2 ms, the CGDT runs well below the standard 2–4 s SCADA polling window, thwarting deceptive commands before feeder protection relays disconnect hospital generation or violate IEC 60601-1 safety margins.

Limitations and Future Directions. While validated under pandapower AC power flow on IEEE 14-bus and 30-bus benchmarks, future work will deploy the CGDT onto embedded RTU microcontrollers and hardware-in-the-loop (HIL) digital simulators (e.g., OPAL-RT) to measure real-time communication

delays and packet jitter. Subsequent efforts will also integrate spatio-temporal graph attention to detect slow, multi-stage attack trajectories.

Acknowledgments. This research was supported by the Research Council of Norway through "Strengthening Resilience in Critical Sectors through IT-OT Integration and Human-Organizational Aspects (ResCri)", Project No. 359829.

Declaration on the Use of AI and AI-Assisted Technologies. AI-assisted tools were used to improve language quality (including spelling, grammar, and punctuation), and all outputs were reviewed, revised, and approved by the authors, who take full responsibility for the final content.

References

1. Ayad, A., Farag, H.E.Z., Youssef, A., El-Saadany, E.F.: Detection of false data injection attacks in smart grids using recurrent neural networks. In: Proc. IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT). pp. 1–5 (2018). https://doi.org/10.1109/ISGT.2018.8403355
2. Barreto, N.E.M., Aoki, A.R.: Cyber-physical power system digital twins—a study on the state of the art. Energies **18**(22), 5960 (2025)
3. Beg, O.A., Johnson, T.T., Davoudi, A.: Detection of false-data injection attacks in cyber-physical DC microgrids. IEEE Transactions on Industrial Informatics **16**(5), 3518–3528 (2020). https://doi.org/10.1109/TII.2017.2656905
4. Boyaci, O., Narimani, M.R., Davis, K., Ismail, M., Overbye, T.J., Serpedin, E.: Joint detection and localization of stealth false data injection attacks in smart grids using graph neural networks. IEEE Transactions on Smart Grid **13**(1), 807–819 (2022)
5. Chockalingam, S., Pieters, W., Teixeira, A., van Gelder, P.: Bayesian network models in cyber security: A systematic review. In: Proc. Nordic Conference on Secure IT Systems (NordSec). Lecture Notes in Computer Science, vol. 10674, pp. 104–120. Springer (2017). https://doi.org/10.1007/978-3-319-70290-2_7
6. Coventry, L., Branley, D.: Cybersecurity in healthcare: A narrative review of trends, threats and ways forward. Maturitas **113**, 48–52 (2018)
7. Deng, R., Zhuang, P., Liang, H.: False data injection attacks against state estimation in power distribution systems. IEEE Transactions on Smart Grid **10**(3), 2871–2881 (2019). https://doi.org/10.1109/TSG.2018.2813280
8. Grieves, M., Vickers, J.: Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. Transdisciplinary Perspectives on Complex Systems pp. 85–113 (2017)
9. Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: Proc. Advances in Neural Information Processing Systems (NeurIPS). pp. 1024–1034 (2017)
10. He, Y., Mendis, G.J., Wei, J.: Real-time detection of false data injection attacks in smart grid: A deep learning-based intelligent mechanism. IEEE Transactions on Smart Grid **8**(5), 2505–2516 (2017). https://doi.org/10.1109/TSG.2017.2703842
11. Kamel Boulos, M.N., Zhang, P.: Digital twins: From personalised medicine to precision public health. Journal of Personalized Medicine **11**(8), 745 (2021)
12. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: Proc. Int. Conf. Learning Representations (ICLR) (2017)

13. Kruse, C.S., Frederick, B., Jacobson, T., Monticone, D.K.: Cybersecurity in healthcare: A systematic review of modern threats and trends. Technology and Health Care **25**(1), 1–10 (2017)
14. Liang, G., Zhao, J., Luo, F., Weller, S.R., Dong, Z.Y.: A review of false data injection attacks against modern power systems. IEEE Transactions on Smart Grid **8**(4), 1630–1638 (2017)
15. Liu, Y., Ning, P., Reiter, M.K.: False data injection attacks against state estimation in electric power grids. ACM Transactions on Information and System Security **14**(1), 1–33 (2011)
16. Luo, X., Li, X., Wang, X., Chan, X., Guan, X.: Detection of false data injection attacks in smart grid: A secure federated deep learning approach. IEEE Transactions on Smart Grid **13**(6), 4862–4872 (2022)
17. Newaz, A.K.M.I., Sikder, A.K., Rahman, M.A., Uluagac, A.S.: A survey on security and privacy issues in modern healthcare systems: Attacks and defenses. ACM Transactions on Computing for Healthcare **2**(3), 1–44 (2021). https://doi.org/10.1145/3453176
18. Pirbhulal, S., Pombo, N., Felizardo, V., Garcia, N., Sodhro, A.H., Mukhopadhyay, S.C.: Towards machine learning enabled security framework for IoT-based healthcare system. In: Proc. 13th International Conference on Sensing Technology (ICST). pp. 1–6 (2019). https://doi.org/10.1109/ICST46873.2019.9047745
19. Sahoo, S., Mishra, S., Peng, J.C.H., Dragicevic, T.: A stealth cyber-attack detection strategy for DC microgrids. IEEE Transactions on Power Electronics **34**(8), 8162–8174 (2019)
20. Sundararajan, M., Taly, A., Yan, Q.: Axiomatic attribution for deep networks. In: Proc. Int. Conf. Machine Learning (ICML). pp. 3319–3328 (2017)
21. Sweeten, J., Elshazly, A., Takiddin, A., Ismail, M., Refaat, S.S., Atat, R.: Cyber-physical fusion for GNN-based attack detection in smart power grids. IEEE Open Access Journal of Power and Energy **12**, 288–299 (2025)
22. Takiddin, A., Ismail, M., Serpedin, E.: Generalized graph neural network-based detection of false data injection attacks in smart grids. IEEE Transactions on Emerging Topics in Computational Intelligence **7**(2), 618–629 (2023)
23. Tao, F., Zhang, H., Liu, A., Nee, A.Y.C.: Digital twin in industry: State-of-the-art. IEEE Transactions on Industrial Informatics **15**(4), 2405–2415 (2019)
24. Thurner, L., Scheidler, A., Schäfer, F., Menke, J.H., Dollichon, J., Meier, F., Meinecke, S., Braun, M.: pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems. IEEE Transactions on Power Systems **33**(6), 6510–6521 (2018)
25. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: Proc. Int. Conf. Learning Representations (ICLR) (2018)
26. Ying, R., Bourgeois, D., You, J., Zitnik, M., Leskovec, J.: GNNExplainer: Generating explanations for graph neural networks. In: Proc. Advances in Neural Information Processing Systems (NeurIPS). pp. 9244–9255 (2019)
27. Yuan, H., Yu, H., Gui, S., Ji, S.: Explainability in graph neural networks: A taxonomic survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(5), 5782–5799 (2023)
28. Yue, X., Qu, G.: Graph neural networks approach for cyberattack detection in smart grids. In: Proc. Cyber Awareness and Research Symposium (CARS). pp. 1–6 (2024)